

Skills Summary

Solving and Graphing One-Step Inequalities

Use this reference while completing your worksheet or when studying.

Undoing an Addition or a Subtraction

Rule: add or subtract the same number on both sides. This *never* changes the direction of the sign.

Graphing: put the circle on the boundary number — **open** for $<$ and $>$, **filled** for $\leq$ and $\geq$ — then shade the side the finished sign points to when the letter is written first.

Example: Solve and graph $b + 7 \leq 10$.

Step 1. The 7 is added to b , so undo it by subtracting 7 from both sides; the sign stays as it is.

Step 2. $b \leq 10 - 7$, that is $b \leq 3$. Graph: filled circle at 3 (because of the bar under the sign), shaded to the left.

Try it: Solve $c - 5 > 2$ and say which way the graph shades.

check: $c > 7$, open circle, shaded right

Dividing or Multiplying by a Positive Number

Rule: divide (or multiply) both sides by the same *positive* number. The direction of the sign stays exactly as it was — even when the answer turns out negative. A negative answer is not a reason to turn the sign around.

Example: Solve and graph $6h < 42$.

Step 1. The letter is multiplied by 6, so divide both sides by 6; six is positive, so the sign keeps its direction.

Step 2. $h < 42 \div 6$, that is $h < 7$. Graph: open circle at 7, shaded to the left.

Try it: Solve $\frac{u}{5} \geq 3$ and say which way the graph shades.

check: $u \geq 15$, filled circle, shaded right

Dividing or Multiplying by a Negative Number

Rule: dividing or multiplying both sides by a *negative* number reverses the order of the two sides, so the sign must **turn around**: $<$ becomes $>$, $\leq$ becomes $\geq$, and back again.

Why: $2 < 5$ is true, but multiplying both sides by -1 gives -2 and -5 , and -2 is the *larger* of those two.

Example: Solve and graph $-4z > 12$.

Step 1. The letter is multiplied by -4 , so divide both sides by -4 — and because that divisor is negative, turn the sign around.

Step 2. $z < 12 \div (-4)$, that is $z < -3$. Graph: open circle at -3 , shaded to the left. Test $z = -4$: $-4 \times (-4) = 16 > 12$ ✓

Try it: Solve $-\frac{w}{2} \leq 5$ and say which way the graph shades.

check: $10 \geq -5$, filled circle, shaded right

(!) Watch out: turn the sign around only when the number you divide or multiply *by* is negative. In $7w < -21$ the divisor is $+7$, so the sign stays and the answer is $w < -3$.